Sushmetha A Mohan

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Current location: +919629304586 India; Relocating to Germany (Metzingen 72555) by Feb 2026 |

EXECUTIVE SUMMARY

Biological researcher with expertise in neuroscience and oncology, focused on translating mechanistic insights into therapeutic discovery. Experienced in primary and PDX-derived cell models, advanced molecular and immunoassays, and in vivo functional studies in transgenic mouse models. My strengths lie in designing, optimizing, and interpreting complex experiments to elucidate mechanisms of action, driving impactful preclinical and translational research in collaborative, fast-paced environments.

EDUCATION

Ph.D in Neuroscience, **McGill University, Canada**	**(GPA: 3.92/4)**	**2017-2023**
B.Tech in Biotechnology, **SASTRA University**	**(GPA: 8.35/10)**	**2012-2016**

TECHNICAL SKILLSET

Molecular biology and Biochemistry

- Nucleic Acid Assays (DNA and RNA Extraction, PCR/qRT-PCR, Agarose Gel Electrophoresis)
- Protein Assays (SDS-PAGE, Immunoblot, Immunoprecipitation (IP), CHIP, RIP, UV-visible spectrometry, BCA assay)
- Molecular cloning and microbiology: DNA ligation, bacterial transformation, In-Fusion cloning, plasmid preparation (mini/maxiprep)
- Histology and imaging: immunohistochemistry, immunofluorescence (cells and tissues), H&E staining

Cell-based Assays:

- Mammalian cell culture: Immortalised cell lines, Primary neural progenitors/neurons, patient-derived brain tumor stem cells, adherent and suspension lines; immune cell isolation and culture (from tumors, spleen)
- *In vivo* skills (in mouse models): Orthotopic and Patient-derived xenograft implantation, intracerebellar, subcutaneous and Intraperitoneal injections, *in-utero* electroporation, Transcardial Perfusion
- Gene manipulation: siRNA, shRNA, CRISPR-Cas9, lentiviral transduction
- Functional assays: cell viability (MTS, Alamar Blue), Dose-response assays (EC50, IC50 curves), dynamic light scattering
- Bulk RNA-seq sample preparation and downstream data analysis

Immunology Assays:

- Flow cytometry, Luminex Multiplex assays

Microscopy:

- Live/fixed cell imaging (Fluorescence & Confocal microscopy) & Microscopy Image Analysis Software (ImageJ & Imaris)

Computer skills/ Bioinformatics:

- Statistical tools (MS-office tools, GraphPad prism), Primer design & analysis, Affinity software, Benchling for lab notebook maintenance
- Datamining and interpretation (using R2 Genomics Analysis and Visualization platform, STRING proteomic database, cBioPortal for Cancer Genomics), Functional annotations using DAVID Bioinformatics and Panther gene ontology

Soft skills:

- Maintain meticulous documentation in compliance with SOPs, GLP/GCLP, and other regulatory guidelines.
- Troubleshooted rigorous protocols for tumor stem cell culturing, virus production, and primary neural stem cell isolation. Shared these with trainees and colleagues to foster training and standardization of procedures.
- Supervised and trained 4 undergraduate and 2 graduate research trainees in molecular assays resulting in successful targeting outcomes
- Served as a teaching mentor for biology and coordination mentor for improving education reforms at iB Hubs mentor program based in India virtually for over 5 years.
- Recipient of departmental, institutional and governmental scholarships during every academic year of my doctoral studies.

RELEVANT EXPERIENCE

Immuni T, Montreal, Canada **Aug 2024- Oct 2025**

Scientific Analyst → Study director

- Progressed into Study Director role with increasing scientific, operational, and regulatory leadership.
- Served as technical and scientific lead for pre-clinical and clinical immunology studies in GLP-compliant environments.
- Performed and oversaw PBMC isolation; primary immune cell and cell line culture; cellular phenotyping and functional assays.
- Executed, optimized, and troubleshot immunological platforms including Luminex, and flow cytometry.
- Led study feasibility assessments, resource planning, timelines, and data analysis to ensure on-time delivery.
- Developed, validated, standardized, and transferred bioanalytical methods in compliance with GLP and SOPs.
- Authored regulatory-ready scientific reports, including U.S. FDA submissions via sponsors, prepared data visualizations and technical documentation.
- Mentored associate analysts and supported laboratory operations, documentation, data integrity, and continuous GLP compliance.
- Worked collaboratively in fast-paced, cross-functional teams within BSL-2 laboratory settings.

Montreal Clinical Research Institute (IRCM), Montreal, Canada **Nov 2023- Dec 2024**
Post-doctoral researcher; Supervisor: Dr. Frederic Charron
To identify, characterize and develop novel therapeutics against pediatric brain tumors (medulloblastoma) using multiple mouse models of this disease , with a specific focus of RNA surveillance pathways in the cell of origin of medulloblastoma

McGill University, Montreal, Canada **2017-2023**
Ph.D student; Supervisor: Dr. Frederic Charron
The nonsense-mediated mRNA decay pathway modulates Sonic hedgehog signaling and medulloblastoma aggressiveness

- Identified and characterized novel tumor suppressors of SHH-MB using multiple tumor mouse models (using bulk RNA-seq sample preparation and data interpretation to identify spontaneous mutations)
- Evaluated the role of NMD in neurodevelopment and medulloblastoma both *in vitro* (using loss- and gain-of function approaches in primary cGNP and tumor cell cultures, nucleic acid and protein assays, microscopy) and *in vivo* (using in-utero electroporation, orthotopic transplantation, H&E staining)
- Identified the mechanism of NMD-mediated modulation of Sonic hedgehog signaling (transcriptional and translational reporter assays, nucleic acid & protein assays, genome-wide bioinformatics to predict mRNA features)
- Maintenance of laboratory notebook and data presentation in weekly meetings

Harvard-MIT, Boston, MA, USA **2016**
Research Assistant; Mentors: Dr. Ashish.A.Kulkarni & Dr. Shiladitya Sengupta
Engineering supramolecular therapeutics for immune modulation in the tumor landscape for improved anti-cancer effect

- Synthesized (using lipid-film hydration), characterized (using Dynamic Light Scattering & spectrophotometry) of histone deacetylation (HDAC) inhibiting lipid based supramolecular nanoparticle (SNP) and tested in mouse models *in vivo* (using subcutaneous dosing and injections)
- Investigated effect of the SNP on neuroblastoma tumor cell proliferation (using biochemistry and microscopy)
- Flow cytometry (surface marker analysis) to quantify skewing of subsets of T-cell populations after HDAC inhibition

SASTRA University, TN, India **2015**
Undergraduate Researcher; Mentors: Dr. N. Sai Subramanian & Dr. Thiagarajan Raman
Screening of nanoparticles for major antibiotic resistance mechanisms in multidrug resistant (MDR) bacterial strains.

- Evaluate antibiofilm activity of copper sulphate nanoparticles in MDR P. aeruginosa strains (using crystal violet and tissue culture plate biofilm assay, plaque formation assays)
- Evaluate impact of copper sulphate nanoparticles on efflux pump genes expression in MDR P. aeruginosa isolates (characterization of nanoparticles using Dynamic Light Scattering, gene expression by qRT-PCR)

SASTRA University, TN, India **2013-2014**
Undergraduate Researcher; Mentors: Dr. Meera. P & Dr. Brindha. P
Hierarchical nanofeatures promote microbial adhesion in tropical grasses: Nanotechnology behind traditional disinfection.

- Evaluate the antibacterial activity, chemical composition (phytochemical methods), hydrophobicity, morphology and nanofeatures of various tropical grasses (assisted with bacterial fixation for Scanning Electron Microscopy).

PUBLICATIONS

Peer-reviewed:

- **Sushmetha A. Mohan**, Chia-Lun Wu, Francois Depault, Jean-Francois Michaud, Sara Calabretta, Amandine Bemmo & Frederic Charron. The nonsense-mediated mRNA decay (NMD) pathway modulates Sonic hedgehog signaling and medulloblastoma aggressiveness. **First author manuscript currently in revision at *Developmental Cell.***
- Lukas Tamayo-Orrego, David Gallo, Frederic Racicot, Amandine Bemmo, **Sushmetha Mohan**, Brandon Ho, Samer Salameh, Trang Hoang, Andrew P. Jackson, Grant W. Brown & Frederic Charron. Sonic hedgehog accelerates DNA replication to cause replication stress promoting cancer initiation in medulloblastoma. *Nat Cancer* 1, 840–854 (2020)
- Deena Santhana Raj, **A.M.Sushmetha**, S. Jayashree, Shrividhya Seshadri, Saritha Balasubramani, Brindha Pemaiah & Meera Parthasarathy. Hierarchical Nanofeatures Promote Microbial Adhesion in Tropical Grasses: Nanotechnology Behind Traditional Disinfection. BioNanoSci 5: 75-83 (2015)

Other publications:

- **S. Mohan**, A. Bonni, and A. Jahani-Asl. "Targeting OSMR in Glioma Stem Cells." Oncotarget 8.10 (2017): 16103–16104
- **S. Mohan** and F. Charron: Faculty Opinions Recommendation of [Cho H et al., Mol Cell 2022]. In Faculty Opinions, 24 Jun 2022; 10.3410/f.742184533.793593741
- **S. Mohan** and F. Charron: Faculty Opinions Recommendation of [Dang MT et al., Cell Rep 2021 34(13):108917]. In Faculty Opinions, 24 Jun 2022; 10.3410/f.739855979.793593742

PRESENTATIONS & POSTERS

- **S.Mohan,** C.Wu, S.Calabretta, A.Bemmo, J-F.Michaud, F.Depault, **F.Charron.** The nonsense-mediated mRNA decay pathway: A new mechanism regulating the proliferation of cerebellum progenitors and its impact on medulloblastoma formation. Canadian Association for Neuroscience - Neural stem cells satellite talk, Canada, 2023.
- **S.Mohan,** C.Wu, S.Calabretta, A.Bemmo, J-F.Michaud, F.Depault, **F.Charron.** Intrinsic regulators of canonical and non-canonical Shh signalling. EMBO workshop: Hedgehog signalling, Spain, 2023.
- **S.Mohan**, C.Wu, A.Bemmo, J-F.Michaud, F.Depault, F.Charron. Nonsense-mediated mRNA decay is a novel suppressor of medulloblastoma formation. IRCM Scientific Forum 2020 and 2022, Canada (Awarded *"Best Oral Presentation Award"*)
- **S.Mohan**, C.Wu, A.Bemmo, F.Charron. Investigation of a novel tumor suppressor mechanism in medulloblastoma. IPN Retreat 2020, BTRSS 2020 and Developmental Biology Seminar Series 2020, McGill University, Canada (Invited for oral presentation)
- **S.Mohan**, V.Chandrasekar, S.V.Natarajan, A.Kulkarni, S.Sengupta. Engineering supramolecular therapeutics for immune modulation in the tumor landscape. SASTRA Unviersity and Harvard-MIT HST 2016, USA (Oral presentation)

REFERENCES

- **Dr. Frederic Charron** - Full Research Professor, Department of Medicine, Université de Montréal / Director, Molecular Biology of Neural Development Research Unit, IRCM, Canada (frederic.charron@ircm.qc.ca)
- **Dr. Saisubramanian Nagarajan** – Assistant Professor: Grade III, SASTRA University , India (sai@scbt.sastra.edu)
- **Dr. Srinivasan Vedantham** - Executive Vice President (Global Operations), DifGen Pharmaceuticals and Aveva Drug Delivery Systems, USA (srinivasan.vedantham@gmail.com)
- **Dr**. **Dhara Patel** – Life sciences marketing lead, Supreme Group (Former employee and my senior manager at ImmuniT) (dharapatel827@gmail.com)